

NATIONAL UNIVERSITY
of Computer & Emerging Sciences, Lahore

Project Phase 2 (Track C)

Memory Priority Scheduling Algorithm for Cloud Data Center Based on Machine Learning Dynamic Clustering Algorithm

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Course: Artificial Intelligence (AI-2002)

Instructor: Ms. Hajra Waheed

Group ID: 3

Group Members:

Mustafa Salman	24L-3051
Mahrukh Rehman	24L-3037
Bilal Ahmed	24L-3090
Rubia Tahir	24L-3098

1. Dataset Source & Description

Source: Alibaba Cluster Trace v2018, published in the public alibaba/clusterdata GitHub repository (cluster-trace-v2018). The paper names this trace as the dataset it first evaluated PMPD on. Because the trace is public, no proxy dataset is needed.

Table used: batch_instance. Each row is one execution of a batch task instance on a machine, which maps naturally to a VM request in the PMPD scheduling model. The raw CSV has no header, so the schema was assigned manually.

Column	Description
instance_name	Unique instance ID
task_name, job_name	Task and job the instance belongs to
task_type	Task category code
status	Terminated, Running, Failed, or Interrupted
start_time, end_time	Instance start and end (seconds from trace start)
machine_id	Machine the instance ran on
seq_no, total_seq_no	Sequence number and total sequence count of the instance
cpu_avg, cpu_max	Average and peak CPU usage
mem_avg, mem_max	Average and peak normalized memory usage

Collection method: The trace was downloaded using the repository's fetchData.sh as batch_instance.tar.gz. Because the full file is very large, only the first 20% of archive members were extracted. The first 1,000,000 rows of the extracted CSV were then loaded with pandas.

2. Brief EDA (Key Insights)

Preprocessing performed

- Assigned column names to the headerless CSV.
- Cast start_time and end_time to integers.
- Derived duration = end_time − start_time.
- Removed rows with negative duration (invalid records).
- Drew a balanced sample of 600 rows per status (seed 42) for visualization.

Key insights

1. **Severe class imbalance.** Of 1,000,000 loaded rows, 991,565 (99.2%) are Terminated; Running has 7,467, Failed 902, Interrupted 66. Analysis on raw data is dominated by Terminated instances, so the visualization uses a balanced sample.
2. **Few invalid records.** The duration filter removed 334 rows (0.03%): Running 242, Failed 26, Interrupted 66. All Interrupted rows were removed, so the cleaned data has three statuses.
3. **Failed instances lack resource metrics.** In the sampled Failed rows, cpu_avg, cpu_max, mem_avg and mem_max are all NaN. These rows cannot supply CPU/memory demand to the scheduler and should be excluded or handled separately.
4. **Durations vary widely.** The Plotly chart of duration by status (balanced sample, with range slider) shows runtimes spanning seconds to several minutes in the rows inspected. This variation matters for PMPD's remaining-time matching rule.
5. **Machine IDs are spread across the cluster** (e.g., m_993, m_1910, m_3430, m_3846), so the sample touches many machines.

Limitations

- The 1,000,000 rows are the first rows of the first 20% of archive members, not a random sample, and may be biased toward early trace periods.
- The balanced 1,800-row sample is for visualization only and does not reflect the true status mix.

3. Final Dataset Summary

Item	Value
Source table	batch_instance (Alibaba Cluster Trace v2018)
Extraction	First 20% of archive members; first 1,000,000 rows
Rows after cleaning	999,666
Rows removed	334 (negative duration)
Columns	14 original + 1 derived (duration) = 15
Status distribution (cleaned)	Terminated 991,565; Running 7,225; Failed 876
Visualization sample	1,800 rows (600 per status, seed 42)
Known missing values	CPU/memory metrics missing for Failed instances (sampled rows)